

IMAT MOCK EXAM 4 — "Calculation Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 43/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Osmosis is the net movement of water across a selectively permeable membrane from a region of lower solute concentration to a region of higher solute concentration. In animal cells, an extracellular solution that is more concentrated than the cytoplasm causes the cell to lose water and shrink; a more dilute solution causes water to enter and the cell to swell."

Which is consistent with the passage?

- A) In a dilute external solution, an animal cell shrinks.
- B) Osmosis requires ATP.
- C) Water moves from high to low solute concentration.
- D) Water moves toward the higher-solute side.
- E) A concentrated external solution causes animal cells to swell.

Answer: D

The passage states that osmosis moves water from lower to higher solute concentration, so the consistent statement is that water moves toward the higher-solute side. The claim that water moves from high to low solute concentration reverses the stated direction, and the claim that a dilute external solution makes an animal cell shrink reverses the stated effect: dilute outside means water enters and the cell swells. Osmosis is passive, so the ATP claim also contradicts the passage.

2. "In genetics, a test cross is used to determine whether an individual displaying a dominant phenotype is homozygous or heterozygous. The individual is crossed with a homozygous recessive partner. If any offspring show the recessive phenotype, the tested individual must be heterozygous."

Which is the strongest inference?

- A) Test crosses require a homozygous dominant partner.
- B) A test cross always produces a 3:1 ratio.
- C) If all offspring show the dominant phenotype, the tested individual is definitely homozygous recessive.
- D) Test crosses can reveal the genotype only of recessive individuals.
- E) Observation of any recessive offspring proves the tested individual is heterozygous.

Answer: E

A recessive offspring must receive a recessive allele from each parent, so a dominant-phenotype parent that produces one must carry the recessive allele, meaning it is heterozygous; this direct restatement of the passage is the strongest inference. The 3:1 ratio belongs to a self-cross of two heterozygotes, not to a test cross (heterozygote × recessive gives roughly 1:1). The claim that all-dominant offspring prove the parent is homozygous recessive is self-contradictory, and the test-cross partner is homozygous recessive, not dominant.

3. (General knowledge) The Italian chemist who proposed a number equal to the number of particles in one mole ($\sim 6.022 \times 10^{23}$), honoured in the constant's name though he did not measure it, is:

- A) Giulio Natta
- B) Amedeo Avogadro
- C) Enrico Fermi
- D) Stanislao Cannizzaro
- E) Mario Molina

Answer: B

The constant is Avogadro's number, $\approx 6.022 \times 10^{23}$ particles per mole, named after the Italian chemist Amedeo Avogadro, whose 1811 hypothesis that equal gas volumes contain equal numbers of molecules underpins the mole concept. Cannizzaro later championed Avogadro's ideas but the constant does not bear his name, and Molina, Fermi and Natta are 20th-century scientists unrelated to the mole.

4. Six boxes are stacked. Box R is above S but below T. Box U is immediately below S. Box V is at the very top. The order from top to bottom could be:

- A) T, V, R, S, U, W
- B) V, U, S, R, T, W
- C) V, T, R, S, U, (and W)
- D) V, R, T, S, U, W
- E) V, T, S, R, U, W

Answer: C

Givens: six boxes R, S, T, U, V, W in one stack. V is at the very top. R is above S and below T, so the order T above R above S holds. U is immediately below S.

Deductions:

V locks position 1.

T, R, S, U form a fixed run of four consecutive slots, because U must sit directly under S at the end of the chain.

The stack has exactly four free slots (positions 2 to 5), so the run fills them in order: position 2 T, 3 R, 4 S, 5 U.

The only box left, W, takes position 6.

Conclusion: from top to bottom the order is V, T, R, S, U, W, and it is unique.

The tempting V, T, S, R, U, W swaps R and S, breaking the rule that R sits above S.

5. All squares are rectangles. Some rectangles are rhombi. Therefore:

- A)** No valid conclusion about squares being rhombi follows.
- B)** No square is a rhombus.
- C)** Some rectangles are squares.
- D)** All squares are rhombi.
- E)** Some squares are rhombi.

Answer: A

Givens: all squares are rectangles; some rectangles are rhombi.

Deductions:

The rectangles that are rhombi are unspecified, so they may all be rectangles other than squares.

No premise forces any square to be a rhombus, and none forbids it either; both 'some squares are rhombi' and 'no square is a rhombus' remain possible.

Conclusion: no valid conclusion about squares being rhombi follows from these premises.

The tempting 'some rectangles are squares' would need squares to exist, which a universal statement alone never guarantees; the everyday fact that a square is a rhombus is outside knowledge, not a deduction from the givens.

6. If 5 machines make 5 widgets in 5 minutes, how long does it take 100 machines to make 100 widgets?

- A)** 1 minute
- B)** 5 minutes
- C)** 20 minutes
- D)** 100 minutes
- E)** 500 minutes

Answer: B

Given: 5 machines make 5 widgets in 5 minutes.

Rate: per machine, $5 \text{ widgets} \div 5 \text{ machines} = 1 \text{ widget per machine in 5 minutes}$.

Substitute: $100 \text{ machines} \times 1 \text{ widget per machine}$, all working in parallel.

100 machines turn out 100 widgets in the same 5 minutes.

Result: 5 minutes.

The tempting 100 minutes scales the time with the machine count; but machines and widgets both grow $\times 20$, so the time per machine never changes.

7. What is the angle between the hands of a clock at 12:30?

- A)** 0°
- B)** 15°
- C)** 30°
- D)** 165°
- E)** 180°

Answer: D

Given: clock reading 12:30.

Relationships: the minute hand covers 360° in 60 min (6° per min); the hour hand covers 360° in 12 h, i.e. 30° per hour.

Minute hand: $30 \text{ min} \times 6^\circ/\text{min} = 180^\circ$, pointing at 6.

Hour hand: $30^\circ \times 0.5 = 15^\circ$ past the 12

Angle between the hands: $180^\circ - 15^\circ = 165^\circ$

Result: 165° .

The tempting 180° treats the hour hand as still exactly on 12 and ignores its 15° drift toward 1.

8. Pointing to a photograph, Maria says, "Her mother is my mother's only daughter." How is the person in the photograph related to Maria?

- A) Maria herself
- B) Maria's mother
- C) Maria's daughter
- D) Maria's aunt
- E) Maria's sister

Answer: C

Givens: Maria says the pictured person's mother is my mother's only daughter.

Deductions:

Maria's mother has only one daughter, so 'my mother's only daughter' is Maria herself.

The pictured person's mother is therefore Maria.

A person whose mother is Maria is Maria's child.

Conclusion: the person in the photograph is Maria's daughter.

The tempting 'Maria herself' fails because the pictured person's mother would then have to be Maria's mother, not Maria.

9. A number is doubled, then 8 is added; the result is 20. The number is:

- A) 4 B) 6 C) 8 D) 10 E) 12

Answer: B

Given: a number x is doubled, then 8 is added, giving 20.

Equation: $2x + 8 = 20$

Undo the adding first: $2x = 20 - 8 = 12$

Undo the doubling: $x = 12 \div 2 = 6$

Check: $2 \times 6 + 8 = 12 + 8 = 20$

Result: $x = 6$

The tempting 10 comes from halving 20 straight away; the reversal must run backwards, subtract 8 first, then halve.

Section B - Biology (Q10-32)

10. The mitochondrial matrix is the site of:

- A) Lactate fermentation
- B) Glycolysis
- C) The light reactions of photosynthesis
- D) The electron transport chain
- E) The Krebs cycle

Answer: E

The matrix is the innermost compartment of the mitochondrion and contains the Krebs (citric acid) cycle enzymes. Glycolysis runs in the cytosol, the electron transport chain is embedded in the inner membrane rather than the matrix, lactate fermentation is cytosolic, and the light reactions belong to chloroplast thylakoids, not mitochondria.

11. The net reaction of cellular respiration is:

- A) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$
- B) $C_6H_{12}O_6 \rightarrow 2 \text{ lactate}$
- C) $C_6H_{12}O_6 \rightarrow 2 \text{ ethanol} + 2CO_2$
- D) $6O_2 + 6CO_2 \rightarrow C_6H_{12}O_6$
- E) $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Answer: A

Aerobic respiration consumes one glucose with six oxygen molecules and yields six CO_2 , six H_2O and energy: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$. The equation $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ runs the other way and is photosynthesis, while $\text{glucose} \rightarrow 2 \text{ lactate}$ and $\text{glucose} \rightarrow 2 \text{ ethanol} + 2CO_2$ are only the anaerobic fermentation branches, not the overall aerobic reaction.

12. Which gas is produced by muscle cells during anaerobic respiration in yeast but NOT in humans?

- A) CO_2 B) CH_4 C) N_2 D) O_2 E) H_2

Answer: A

Yeast ferments pyruvate to ethanol plus CO_2 , so it releases carbon dioxide; human muscle reduces pyruvate to lactate and releases no CO_2 . CO_2 is therefore produced in yeast anaerobic respiration but not in human muscle. O_2 , N_2 , CH_4 and H_2 are products of neither pathway.

13. The inner mitochondrial membrane is folded into:

- A) Villi B) Microvilli C) Thylakoids D) Cisternae E) Cristae

Answer: E

The inner mitochondrial membrane folds into cristae, which enlarge the surface available for electron transport chain complexes and ATP synthase. Thylakoids are chloroplast membranes, cisternae are flattened Golgi/ER sacs, and microvilli and villi are folds of cell surfaces or the intestine, none of them mitochondrial.

14. The Calvin cycle of photosynthesis occurs in the:

- A) Cytosol
- B) Thylakoid membrane
- C) Matrix
- D) Stroma
- E) Nucleus

Answer: D

The Calvin cycle, i.e. the light-independent fixation of CO_2 , takes place in the stroma, the fluid surrounding the thylakoids. The thylakoid membrane hosts the light-dependent reactions instead, and the matrix is a mitochondrial compartment, so neither fits.

15. Which is the correct site of the light-dependent reactions?

- A) Stroma
- B) Matrix
- C) Thylakoid membrane
- D) Cell wall
- E) Cytosol

Answer: C

The light-dependent reactions occur on the thylakoid membranes, where the photosystems, the electron transport chain and ATP synthase are located. The stroma instead hosts the Calvin cycle, and the cytosol, matrix and cell wall are not sites of the light reactions.

16. A solution of pH 2 is how many times more acidic than a solution of pH 5?

- A) 3 times B) 10 times C) 30 times D) 100 times E) 1000 times

Answer: E

Given: one solution at pH 2, another at pH 5.

Relationship: each pH unit is a factor of 10 in H^+ concentration, so the acidity ratio is 10 raised to the pH difference.

Substitute: pH difference = $5 - 2 = 3$ units

Ratio = $10^3 = 1000$

Result: the pH 2 solution is 1000 times more acidic.

The tempting 30 treats the pH scale as linear and multiplies the 3-unit gap by 10; the scale is logarithmic, so whole powers of 10 apply.

17. The four levels of protein structure in order are:

- A) Quaternary, tertiary, secondary, primary
B) Primary, tertiary, secondary, quaternary
C) Primary, secondary, quaternary, tertiary
D) Primary, secondary, tertiary, quaternary
E) Secondary, primary, tertiary, quaternary

Answer: D

The levels build up as primary (amino acid sequence), secondary (local α -helix and β -sheet held by backbone hydrogen bonds), tertiary (overall 3D fold of one polypeptide) and quaternary (assembly of several chains). Listings that swap secondary and tertiary or reverse the whole order misplace these levels.

18. Which monosaccharide is the primary energy source for cells and the product of dietary carbohydrate digestion?

- A) Fructose B) Galactose C) Glucose D) Lactose E) Sucrose

Answer: C

Glucose is the monosaccharide used directly in respiration and the main product of dietary carbohydrate (starch and glycogen) digestion. Fructose and galactose are other monosaccharides that must first be converted into glycolytic intermediates, while sucrose and lactose are disaccharides and not monosaccharides at all.

19. Which of the following is a disaccharide?

- A) Sucrose B) Glycogen C) Cellulose D) Fructose E) Glucose

Answer: A

Sucrose consists of glucose joined to fructose by a glycosidic bond, making it a disaccharide. Glucose and fructose themselves are monosaccharides, and cellulose and glycogen are long-chain polysaccharides.

20. Lipids are largely insoluble in water because they are:

- A) Highly polar
- B) Non-polar (hydrophobic)
- C) Composed only of amino acids
- D) Charged
- E) Always small molecules

Answer: B

Lipids are built around long non-polar hydrocarbon chains, so water cannot form favourable interactions with them: they are hydrophobic and hence water-insoluble. Charged or highly polar species dissolve readily, so those claims go the wrong way, and lipids are neither amino-acid based nor necessarily small molecules.

21. DNA differs from RNA in that DNA:

- A) Has fewer nucleotide types
- B) Is found only in the cytoplasm
- C) Contains the sugar ribose
- D) Contains thymine instead of uracil
- E) Is usually single-stranded

Answer: D

The decisive fact is that DNA uses the base thymine where RNA uses uracil. The same contrast rules out the other claims: DNA contains deoxyribose rather than ribose, DNA is usually double-stranded (RNA is the usually single-stranded one), both use four types of nucleotide, and DNA resides chiefly in the nucleus, not the cytoplasm.

22. A population is in Hardy-Weinberg equilibrium for a locus with two alleles A (frequency p) and a (frequency q). The frequency of the heterozygote Aa is:

- A) q^2 B) $2pq$ C) 1 D) $p + q$ E) p^2

Answer: B

In Hardy-Weinberg equilibrium the genotype frequencies are p^2 for AA, $2pq$ for the heterozygote Aa and q^2 for aa, with $p + q = 1$. The heterozygote frequency is therefore $2pq$; p^2 and q^2 are the two homozygotes, and $p + q$ is simply 1.

23. ABO blood groups are an example of:

- A) Polygenic inheritance
- B) Sex-linked inheritance
- C) X-inactivation
- D) Mitochondrial inheritance
- E) Codominance (and multiple alleles)

Answer: E

Type AB individuals express both the A and B antigens, showing codominance, and the system involves three alleles (I^A , I^B , i) in the population, i.e. multiple alleles. The trait is autosomal, so sex-linked inheritance, mitochondrial inheritance, polygenic inheritance and X-inactivation do not describe it.

24. In humans, the normal diploid number of chromosomes is:

- A) 23 B) 44 C) 46 D) 48 E) 92

Answer: C

Human somatic (diploid) cells carry $2n = 46$ chromosomes, arranged as 23 pairs. The value 23 is the haploid (gamete) number, and 44 counts only the 22 autosome pairs while omitting the two sex chromosomes.

25. The cell-cycle phase in which DNA is replicated is:

- A) S phase B) Cytokinesis C) G2 D) M phase E) G1

Answer: A

DNA replication occurs in S (synthesis) phase of interphase, when the genome is copied. G1 and G2 are growth phases without replication, M phase is mitosis, and cytokinesis divides the cytoplasm.

26. Which vessels have thick muscular walls and carry blood away from the heart under high pressure?

- A) Venules B) Arteries C) Veins D) Lymph vessels E) Capillaries

Answer: B

Arteries carry blood away from the heart at high pressure and therefore have thick, elastic, muscular walls. Veins and venules return blood at low pressure through thinner walls, capillaries are one-cell-thick exchange vessels, and lymph vessels are also thin-walled and low pressure.

27. The exchange of O_2 and CO_2 between blood and air occurs by:

- A) Phagocytosis
B) Exocytosis
C) Active transport
D) Osmosis
E) Diffusion (passive)

Answer: E

Gas exchange across the alveolar and capillary walls occurs by passive diffusion down partial-pressure gradients: O_2 moves in and CO_2 out with no energy cost. Osmosis concerns water movement, and active transport, phagocytosis and exocytosis are energy-requiring processes not used for respiratory gas exchange.

28. The hormone secreted by the adrenal medulla in the "fight or flight" response is:

- A) Adrenaline (epinephrine)
B) Thyroxine
C) Insulin
D) Oestrogen
E) Cortisol

Answer: A

The adrenal medulla secretes adrenaline (epinephrine), which drives the fight-or-flight response: raised heart rate, blood pressure and blood glucose. Cortisol is made in the adrenal cortex, insulin in the pancreas, thyroxine in the thyroid and oestrogen in the ovaries, so none of them is the medulla's emergency hormone.

29. The longest bone in the human body is the:

- A) Tibia B) Humerus C) Femur D) Spine E) Skull

Answer: C

The femur, the thigh bone, is the longest and strongest bone in the human body. The tibia and humerus are also long bones but shorter, and the spine and skull are structures made of many bones, not single long bones.

30. Smooth muscle is characterised by:

- A) Multinucleation
B) Presence in the heart only
C) Branched cells with intercalated discs
D) Lack of striation and involuntary control
E) Striation and voluntary control

Answer: D

Smooth muscle consists of spindle-shaped cells with no striations and is controlled involuntarily by the autonomic nervous system. Striations plus voluntary control describe skeletal muscle, and branched cells with intercalated discs describe cardiac muscle, as does presence in the heart only.

31. The organelle responsible for ATP production in animal cells is the:

- A) Mitochondrion B) Nucleus C) Chloroplast D) Lysosome E) Ribosome

Answer: A

Mitochondria perform oxidative phosphorylation and supply most of the ATP in animal cells. Chloroplasts do this job only in plants (animal cells have none), ribosomes build proteins, lysosomes digest material and the nucleus stores DNA, so none of them is the ATP-producing organelle.

32. Which is the correct sequence of blood flow through the heart, starting from the body?

- A) Vena cava → left atrium → left ventricle → aorta
B) Vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta
C) Aorta → right atrium → right ventricle → pulmonary vein
D) Right atrium → left atrium → right ventricle → left ventricle
E) Pulmonary vein → right atrium → right ventricle → pulmonary artery

Answer: B

From the body, blood enters the right side via the vena cava, flows right atrium → right ventricle → pulmonary artery → lungs, returns via the pulmonary vein → left atrium → left ventricle and leaves through the aorta. Shortcuts that start at the aorta or the pulmonary vein, or that route pulmonary-vein blood into the right atrium, contradict this one-way circuit.

Section C - Chemistry (Q33-47)

33. Calculate the number of moles in 22 g of CO₂. (C = 12, O = 16.)

- A) 0.25 B) 0.5 C) 1 D) 2 E) 22

Answer: B

Given: mass $m = 22$ g of CO₂; atomic masses C = 12 g/mol, O = 16 g/mol

Formula: amount $n = \text{mass} \div \text{molar mass}$, $n = m \div M$.

Molar mass: $M(\text{CO}_2) = 12 + (2 \times 16)$

$2 \times 16 = 32$, so $M(\text{CO}_2) = 12 + 32 = 44$ g/mol

Substitute: $n = 22 \text{ g} \div 44 \text{ g/mol}$

$22 \div 44 = 0.5$

Result: $n = 0.5$ mol

The tempting 2 mol comes from inverting the division ($44 \div 22$); moles are mass divided by molar mass, never the reverse.

34. What volume does 0.25 mol of an ideal gas occupy at STP ($V_m = 22.4$ L/mol)?

- A) 2.8 L B) 5.6 L C) 11.2 L D) 22.4 L E) 44.8 L

Answer: B

Given: $n = 0.25$ mol of ideal gas at STP; molar volume $V_m = 22.4$ L/mol.

Formula: volume $V = n \times V_m$.

Substitute: $V = 0.25 \text{ mol} \times 22.4 \text{ L/mol}$

$0.25 \times 22.4 = 5.6$

Result: $V = 5.6$ L

The tempting 11.2 L is the volume of 0.5 mol, twice the given amount; a quarter mole fills a quarter of 22.4 L.

35. How many atoms are in 1.5 mol of He?

- A) 1.20×10^{24} B) 1.50×10^{23} C) 3.01×10^{23} D) 6.02×10^{23} E) 9.03×10^{23}

Answer: E

Given: $n = 1.5$ mol of He; Avogadro's number = 6.022×10^{23} particles per mol

Relationship: atoms = $n \times \text{Avogadro's number}$; helium is monatomic, so each particle is a single atom.

Substitute: atoms = $1.5 \text{ mol} \times 6.022 \times 10^{23} \text{ per mol}$

$1.5 \times 6.022 = 9.033$, so atoms $\approx 9.03 \times 10^{23}$

Result: 9.03×10^{23} atoms.

The tempting 6.02×10^{23} is exactly 1 mol of atoms and drops the factor 1.5; 3.01×10^{23} would be only 0.5 mol.

36. What is the molarity of a solution made by dissolving 5.85 g NaCl (molar mass 58.5) in water to make 500 mL?

- A)** 0.1 M **B)** 0.2 M **C)** 0.5 M **D)** 1.0 M **E)** 2.0 M

Answer: B

Given: $m = 5.85 \text{ g NaCl}$; $M = 58.5 \text{ g/mol}$; final volume 500 mL

Formulas: $n = m \div M$, then concentration $c = n \div V$, with V in litres.

Moles: $n = 5.85 \text{ g} \div 58.5 \text{ g/mol} = 0.1 \text{ mol}$

Volume: $500 \text{ mL} = 0.5 \text{ L}$

Substitute: $c = 0.1 \text{ mol} \div 0.5 \text{ L}$

$0.1 \div 0.5 = 0.2$

Result: $c = 0.2 \text{ M}$

The tempting 0.1 M quotes the mole amount as a concentration and forgets to divide by the 0.5 L volume.

37. The oxidation number of Cr in $\text{Cr}_2\text{O}_7^{2-}$ is:

- A)** -2 **B)** +3 **C)** +6 **D)** +7 **E)** +12

Answer: C

Given: ion $\text{Cr}_2\text{O}_7^{2-}$; oxygen's oxidation number is -2; all oxidation numbers sum to the ion charge, -2.

Equation: $2x + 7 \times (-2) = -2$, with x the oxidation number of one Cr.

Oxygen's total: $7 \times (-2) = -14$

So $2x = -2 + 14 = +12$

$x = 12 \div 2 = +6$

Result: Cr is +6.

The tempting +12 stops at $2x$ and forgets to divide by the two chromium atoms.

38. Identify the oxidising agent in: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$.

- A)** Cu
B) H_2O
C) H_2 (it is oxidised)
D) CuO (it is reduced)
E) None

Answer: D

Copper goes from +2 in CuO to 0 in Cu, so CuO gains electrons and is reduced; the species that is reduced is by definition the oxidising agent. H_2 loses electrons (it is oxidised to H_2O) and is the reducing agent, and the copper metal is the reduction product, not an agent.

39. Calculate the pH of a 0.05 M solution of a strong base NaOH (assume full dissociation, 25 °C).

- A) 1.3 B) 7 C) 12.7 D) 13 E) 14

Answer: C

Given: strong base NaOH, $c = 0.05$ M, full dissociation, 25 °C so $\text{pH} + \text{pOH} = 14$.

Steps: $[\text{OH}^-] = 0.05$ M; $\text{pOH} = -\log[\text{OH}^-]$; $\text{pH} = 14 - \text{pOH}$

Substitute: $\text{pOH} = -\log(0.05) = -\log(5 \times 10^{-2})$

$\log 5 \approx 0.70$, so $\text{pOH} = 2 - 0.70 = 1.30$

Then $\text{pH} = 14 - 1.30 = 12.70$

Result: $\text{pH} \approx 12.7$.

The tempting 1.3 is the pOH misread as the pH; a base must land above 7.

40. The conjugate base of H_2O is:

- A) H^+ B) O_2 C) H_3O^+ D) OH^- E) H_2

Answer: D

A conjugate base is what remains after the acid donates one proton: $\text{H}_2\text{O} - \text{H}^+ = \text{OH}^-$, so the hydroxide ion is water's conjugate base. H_3O^+ is instead water's conjugate acid, formed when water accepts a proton, and H^+ is the transferred particle itself, not the remaining species.

41. Which has the greatest first ionisation energy?

- A) Ar B) Cl C) Na D) Al E) Mg

Answer: A

First ionisation energy rises across Period 3 as nuclear charge pulls on electrons in the same shell, peaking at the noble gas; argon's complete third shell makes it the highest of the five. Sodium is the lowest with its single, easily lost 3s electron, and chlorine, though high, still lies below argon.

42. Which molecule has a linear geometry?

- A) CH_4 B) H_2O C) SO_2 D) NH_3 E) CO_2

Answer: E

CO_2 has two electron domains around the central carbon (two double bonds, no lone pairs), which spread to 180° , giving linear $\text{O}=\text{C}=\text{O}$. H_2O is bent (two lone pairs on O), NH_3 is trigonal pyramidal, SO_2 is bent, and CH_4 is tetrahedral with four bonding pairs, so none of them is linear.

43. Which intermolecular force is strongest?

- A) Hydrogen bonding
B) London dispersion
C) Van der Waals (general)
D) Dipole-dipole
E) Covalent bonding (intramolecular)

Answer: A

Hydrogen bonding, the attraction between H bonded to N, O or F and a lone pair on another electronegative atom, is the strongest of the intermolecular forces listed; dipole-dipole and London dispersion (the van der Waals family) are weaker. Covalent bonding is an intramolecular bond holding atoms within a molecule, so it does not count as an intermolecular force.

44. Increasing pressure on a gas at constant temperature will (Boyle's law):

- A) Increase its volume
- B) Increase its temperature
- C) Decrease its volume
- D) Have no effect
- E) Decrease its moles

Answer: C

Boyle's law, $pV = \text{constant}$ at fixed temperature, makes pressure and volume inversely proportional ($P \propto 1/V$), so increasing pressure decreases the volume. Temperature is constant by assumption and the amount of gas is fixed, so neither the temperature nor the mole number can change.

45. The structural formula $\text{CH}_3\text{-CH}_2\text{-OH}$ represents:

- A) Ethanoic acid
- B) Diethyl ether
- C) Propanol
- D) Ethanol
- E) Methanol

Answer: D

The chain $\text{CH}_3\text{-CH}_2\text{-OH}$ has two carbons bearing a hydroxyl group, which is ethanol ($\text{C}_2\text{H}_5\text{OH}$). Methanol would have one carbon, propanol three, ethanoic acid would carry a -COOH group instead of -OH , and an ether would have C-O-C without the O-H bond.

46. In the reaction: $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$, the precipitate is:

- A) NaCl
- B) None — no precipitate forms
- C) NaNO_3
- D) AgNO_3
- E) AgCl (insoluble)

Answer: E

Mixing the solutions exchanges ions, and silver chloride is insoluble, so AgCl comes out of solution as a white precipitate while NaNO_3 stays dissolved (sodium salts and nitrates are always soluble). The two reactants are the dissolved starting salts, and a precipitate does indeed form.

47. For a reaction at equilibrium, adding a catalyst will:

- A) Stop the reaction
- B) Shift equilibrium toward reactants
- C) Shift equilibrium toward products
- D) Decrease K_{eq}
- E) Increase the rate of both forward and reverse equally; equilibrium position unchanged

Answer: E

A catalyst lowers the activation energy of the forward and reverse reactions by the same amount, so both rates rise equally and equilibrium is simply reached sooner, with the position and K_{eq} unchanged. Any claim of a shift toward products or reactants, or of a changed K_{eq} , contradicts this symmetry, and the reaction does not stop.

Section D - Mathematics (Q48-54)

48. Solve: $2x^2 - 8 = 0$.

- A) No real solution
- B) $x = \pm 4$
- C) $x = \pm 2$
- D) $x = 4$ only
- E) $x = 2$ only

Answer: C

Given: $2x^2 - 8 = 0$

Rearrange: $2x^2 = 8$

Divide by 2: $x^2 = 8 \div 2 = 4$

Square root both sides: $x = +2$ or $x = -2$, because both square to 4.

Result: $x = \pm 2$

The tempting $x = 2$ only drops the negative root; $x = \pm 4$ skips the division by 2 before taking the root.

49. Solve: $|x - 3| = 5$.

- A) $x = 2$
- B) $x = -2$ only
- C) $x = 8$ only
- D) $x = 8$ or $x = -2$
- E) $x = 5$

Answer: D

Given: $|x - 3| = 5$

Meaning: the distance from x to 3 is 5, so the inside equals +5 or -5.

Case +5: $x - 3 = 5$, so $x = 3 + 5 = 8$

Case -5: $x - 3 = -5$, so $x = 3 - 5 = -2$

Result: $x = 8$ or $x = -2$

The tempting single answers (8 only, or -2 only) each keep one case of the absolute value and discard the other.

50. Solve: $\log_2(x) + \log_2(x - 3) = 2$ ($x > 3$).

- A) $x = 4$ B) $x = -1$ C) $x = 3$ D) $x = 8$ E) $x = 1$

Answer: A

Given: $\log_2(x) + \log_2(x - 3) = 2$, with $x > 3$

Combine the logs: $\log_2[x(x - 3)] = 2$

Undo the log: $x(x - 3) = 2^2 = 4$

Expand: $x^2 - 3x - 4 = 0$

Factor: $(x - 4)(x + 1) = 0$, so $x = 4$ or $x = -1$

Apply the domain $x > 3$: $x = -1$ is rejected, $x = 4$ survives

Result: $x = 4$

The tempting $x = -1$ solves the quadratic but makes $\log_2(x - 3)$ undefined, so it is extraneous.

51. Solve the inequality: $x^2 - 4 < 0$.

- A) $x < -2$ B) $x < 2$ C) $x \neq 2$ D) $x > 2$ E) $-2 < x < 2$

Answer: E

Given: $x^2 - 4 < 0$.

Rearrange: $x^2 < 4$, i.e. $(x - 2)(x + 2) < 0$.

Boundaries: the roots $x = -2$ and $x = +2$ split the number line into three zones.

Test the middle, $x = 0$: $0^2 - 4 = -4 < 0$, true; points beyond ± 2 give a positive product.

Result: $-2 < x < 2$.

The tempting ' $x < -2$ or $x > 2$ ' solves the reversed inequality $x^2 > 4$; a square smaller than a number lies between the roots.

52. A right circular cylinder has radius 3 and height 10. What is its volume? (Use $\pi = 3.14$.)

- A) 30 B) 94.2 C) 282.6 D) 314 E) 942

Answer: C

Given: cylinder radius $r = 3$, height $h = 10$, $\pi = 3.14$

Formula: $V = \pi r^2 h$

Substitute: $V = 3.14 \times 3^2 \times 10$

$3^2 = 9$, so $V = 3.14 \times 9 \times 10$

$3.14 \times 9 = 28.26$

$28.26 \times 10 = 282.6$

Result: $V = 282.6$

The tempting 94.2 forgets to square the radius and computes $3.14 \times 3 \times 10$.

53. $\tan 45^\circ = ?$

- A) 1 B) $\sqrt{3}$ C) 0 D) $\sqrt{3}/3$ E) $1/2$

Answer: A

Given: $\tan 45^\circ$.

Method: in a 45-45-90 right triangle the two legs are equal, and $\tan = \text{opposite} \div \text{adjacent}$.

Substitute: $\tan 45^\circ = 1 \div 1$

Result: $\tan 45^\circ = 1$

The tempting $\sqrt{3}$ and $\sqrt{3}/3$ are $\tan 60^\circ$ and $\tan 30^\circ$, dragged in from the 30-60-90 special triangle.

54. A die is rolled once. What is the probability of an even number greater than 2?

- A) $1/6$ B) $1/3$ C) $1/2$ D) $2/3$ E) $5/6$

Answer: B

Given: one fair six-sided die; success = even face and greater than 2.

Sample space: 6 equally likely faces.

Favourable faces: even and above 2 gives 4 and 6, so 2 outcomes.

Substitute: $P = 2 \div 6$

$2 \div 6 = 1/3$

Result: $P = 1/3$

The tempting $1/2$ also counts the face 2, but 2 is even without being greater than 2.

Section E - Physics (Q55-60)

55. A car accelerates from 5 m/s to 25 m/s in 10 s. What is its acceleration?

- A) 1 m/s² B) 2 m/s² C) 5 m/s² D) 20 m/s² E) 25 m/s²

Answer: B

Given: initial speed 5 m/s, final speed 25 m/s, time 10 s.

Formula: acceleration $a = (\text{final speed} - \text{initial speed}) \div \text{time}$.

Substitute: $a = (25 - 5) \div 10$

$25 - 5 = 20$, so $a = 20 \div 10 = 2 \text{ m/s}^2$

Result: $a = 2 \text{ m/s}^2$

The tempting 20 m/s² is the velocity change itself, never divided by the 10 s.

56. A 12 V battery drives a current through a 4 Ω resistor in series with a 2 Ω resistor. What is the current?

- A) 1 A B) 2 A C) 3 A D) 6 A E) 12 A

Answer: B

Given: supply $V = 12 \text{ V}$; resistors 4 Ω and 2 Ω in series

Formulas: series resistances add, $R_{\text{total}} = R_1 + R_2$; then Ohm's law $I = V \div R$.

Substitute: $R_{\text{total}} = 4 \Omega + 2 \Omega = 6 \Omega$

$I = 12 \text{ V} \div 6 \Omega$

$12 \div 6 = 2$

Result: $I = 2 \text{ A}$

The tempting 3 A divides by the 4 Ω alone and ignores the 2 Ω in series with it.

57. A 3 A current flows for 2 minutes. What total charge passes a point?

- A) 6 C B) 60 C C) 180 C D) 360 C E) 600 C

Answer: D

Given: current $I = 3 \text{ A}$ flowing for $t = 2 \text{ minutes}$.

Formula: charge $Q = I \times t$, with t in seconds.

Convert: $t = 2 \times 60 = 120 \text{ s}$

Substitute: $Q = 3 \text{ A} \times 120 \text{ s}$

$3 \times 120 = 360$

Result: $Q = 360 \text{ C}$

The tempting 6 C treats the 2 of '2 minutes' as 2 seconds; charge calculations need time in seconds.

58. A 2 kg object moving at 3 m/s collides with a stationary 1 kg object and they stick together. What is their common velocity?

- A)** 1 m/s **B)** 2 m/s **C)** 3 m/s **D)** 4 m/s **E)** 6 m/s

Answer: B

Given: object of 2 kg moving at 3 m/s hits a stationary 1 kg object; they stick together.

Principle: conservation of momentum, total p before = total p after; after the collision the mass is 2 + 1 = 3 kg.

Substitute: p before = $(2 \times 3) + (1 \times 0) = 6 \text{ kg}\cdot\text{m/s}$

After: $3 \times v = 6$

$v = 6 \div 3 = 2 \text{ m/s}$

Result: $v = 2 \text{ m/s}$

The tempting 6 m/s mistakes the initial momentum value for a velocity; the momentum is carried by all 3 kg.

59. A 4 kg object is dropped (from rest). Ignoring air resistance, what is its velocity after 3 s? ($g = 9.8 \text{ m/s}^2$.)

- A)** 9.8 m/s **B)** 19.6 m/s **C)** 29.4 m/s **D)** 39.2 m/s **E)** 88.2 m/s

Answer: C

Given: object of 4 kg dropped from rest, $t = 3 \text{ s}$, $g = 9.8 \text{ m/s}^2$.

Formula: $v = v_0 + g \times t$, with $v_0 = 0$ for a drop

Substitute: $v = 0 + (9.8 \times 3)$

$9.8 \times 3 = 29.4$

Result: $v = 29.4 \text{ m/s}$

The tempting 39.2 is actually the weight, $4 \text{ kg} \times 9.8 \text{ m/s}^2 = 39.2 \text{ N}$; the mass never affects free-fall speed.

60. A wave has frequency 50 Hz and wavelength 4 m. What is its speed?

- A)** 12.5 m/s **B)** 50 m/s **C)** 100 m/s **D)** 200 m/s **E)** 400 m/s

Answer: D

Given: frequency $f = 50 \text{ Hz}$, wavelength $\lambda = 4 \text{ m}$

Formula: wave speed $v = f \times \lambda$.

Substitute: $v = 50 \text{ Hz} \times 4 \text{ m}$

$50 \times 4 = 200$

Result: $v = 200 \text{ m/s}$

The tempting 12.5 m/s divides 50 by 4 instead of multiplying, inverting the wave equation.